Identity-related Speech Suppression in Generative AI Content Moderation

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ABSTRACT

Automated content moderation has long been used to help identify and filter undesired user-generated content online. But such systems have a history of incorrectly flagging content by and about marginalized identities for removal. Generative AI systems now use such filters to keep undesired generated content from being created by or shown to users. While a lot of focus has been given to making sure such systems do *not* produce undesired outcomes, considerably less attention has been paid to making sure appropriate text *can* be generated. From classrooms to Hollywood, as generative AI is increasingly used for creative or expressive text generation, whose stories will these technologies allow to be told, and whose will they suppress?

In this paper, we define and introduce measures of speech suppression, focusing on speech related to different identity groups incorrectly filtered by a range of content moderation APIs. Using both short-form, user-generated datasets traditional in content moderation and longer generative AI-focused data, including two datasets we introduce in this work, we create a benchmark for measurement of speech suppression for nine identity groups. Across one traditional and four generative AI-focused automated content moderation services tested, we find that identity-related speech is more likely to be incorrectly suppressed than other speech. We find that reasons for incorrect flagging behavior vary by identity based on stereotypes and text associations, with, e.g., disability-related content more likely to be flagged for self-harm or health-related reasons while non-Christian content is more likely to be flagged as violent or hateful. As generative AI systems are increasingly used for creative work, we urge further attention to how this may impact the creation of identity-related content.

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CCS CONCEPTS

• **Computing methodologies** → **Artificial intelligence**; • **Human-centered computing**; • **General and reference** → **Measurement**; • **Applied computing** → *Arts and humanities*;

KEYWORDS

Algorithm audits, generative AI, automated content moderation, fairness in machine learning.

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1 INTRODUCTION

Automated content moderation systems are used across the web to help reduce the occurrence of violent, hateful, sexual, or otherwise undesired user-generated content online, including in online comment sections and by social media platforms [21, 42, 53]. As content is generated by AI systems, automated content moderation techniques are being applied to the text generated by these systems to filter unwanted content before it is shown to users [46, 48]. Such filtering is necessary to help prevent harmful content that generative AI systems are known to produce, including violent, sexual, and hateful content [55]. However, making nuanced determinations about appropriate content is known to be hard, even in traditional contexts where some cases are left to human content moderators [60]. Additionally, in traditional contexts, content moderation is known to suffer from identity-related biases, such that speech by or about marginalized identities is more likely to be incorrectly flagged as inappropriate content [14, 24, 62]. Here, we examine whether and how these biases translate to a generative AI setting. *As generative AI is increasingly used for creative and expressive text generation from schools to Hollywood, this paper is motivated by the question: whose stories won't be told?*

Our focus in this paper on the blocking of speech and the associated use of the term "speech suppression" is in some tension with concerns about whether *lack of* appropriate content moderation of user-generated content and comments will drive marginalized

users from a platform due to, e.g., extensive hate speech. In a generative AI context, there has been extensive work from an AI safety perspective on identifying and filtering out unwanted content, including hate speech, privacy violations, sexual abuse or harassment, and many other concerning or illegal use cases (see [74] for a recent benchmark). The danger we identify and assess here, is that these efforts and the substantial public attention focused on blocking concerning outputs may have the unwanted effect of incorrectly removing desired speech.

We conduct an audit of five automated content moderation systems to measure identity-related speech suppression, introducing benchmark datasets and definitions to quantify these biases in the context of generative AI systems. Previous benchmark datasets for content moderation have focused on user-generated content, such as tweets or internet comments, that have been hand-labeled according to a content moderation rubric [8, 22]. However, most of these datasets are composed of short-form content and do not include the types of text involved in generative AI systems. Here, we focus on the capacity of generative text systems to be used as part of a process of generating expressive speech, especially as part of a storytelling process; we assess whether automated content moderation will mean that stories related to people's demographic backgrounds, experiences, and identities will be automatically removed or kept from being generated in the first place.

We assess content moderation APIs for identity-related speech suppression, focusing on the following experimental research questions:

(1) How much speech suppression do identity groups experience across content moderation APIs?
(2) Do some identity groups experience greater speech suppression than others?
(3) Do APIs perform differently on creative generative AI text than on shorter traditional user-generated text?

1.1 Contributions

This paper makes the following contributions.

Defining the problem. We define identity-related speech suppression in the context of automated content moderation. We provide aggregate measures that assess the extent to which text related to an identity group that should not be filtered is moderated beyond the usual error for non-identity-related text. See Section 3.

Audit and methodology. We introduce a new audit methodology and provide the first comprehensive bias audit of generative AI speech suppression across five automated content moderation APIs (Jigsaw's Perspective, Google Cloud, Anthropic, OctoAI's LLama Guard, and OpenAI's Moderation Endpoint) based on seven datasets and for nine identity groups (women, men, Christian, non-Christian, straight, LGBT, white, non-white, and disability). See Section 4.

Findings. We find that identity-related speech is more likely to be suppressed than other speech for all identity groups, including both marginalized and non-marginalized groups, except two (straight and Christian), with slight variation across APIs. We also find that APIs are better able to avoid incorrect speech suppression on generative AI content, and that TV violence is likely to be suppressed even when normatively considered acceptable (e.g., PG-13 rated movie content). When considering the reasons for identity-related speech suppression, we find that content is suppressed for different reasons across identities based on stereotypes and text associations. For example, disability-related content is flagged as related to self-harm, LGBT and straight content is identified as inappropriately sexual, and non-Christian content is flagged for hate. See Section 5.

Datasets and code. We additionally introduce two new datasets focused on creative content (movies and television shows) with associated content moderation labels and identity-group tags. We create and make public seven benchmark datasets, including our method for tagging text data with nine identity categories (see Sections 4.1 and 4.2). We also open source our a pipeline for running five popular content moderation APIs on this data to assess speech suppression results as we do in this paper. Code and data is available at: https://github.com/genAIaudits/speech-suppression.

Overall, we find that most of the nine identity groups are likely to have their creative speech incorrectly suppressed by a range of popular, commercially-available automated content moderation systems, raising questions about the use of these systems as part of creative generative AI pipelines, and identifying a tension and potential tradeoff between filtering out undesired content and ensuring that other speech is allowed.

2 RELATED WORK

In this paper, we consider whether automated content moderation systems — instituted as part of generative AI systems to avoid various AI harms of concern within the AI safety literature — are biased in their errors, resulting in the suppression of identity-related speech. Thus, we build on work from three key areas: content moderation, generative AI bias audits, and AI safety.

2.1 Content moderation

Content moderation as traditionally performed in online contexts is a partially automated process, where user-generated content such as comments, images, and videos, is automatically or manually identified as content potentially violating a company's policies and may then be examined by a person to determine whether it is removed from the platform based on a platform's policies on violating content [60]. These content moderation jobs are generally low wage and emotionally disturbing work, with contract and "gig" workers spending their days seeing or reading violent, sexual, or otherwise disturbing content [33, 60]. Despite this labor, comment sections and platforms are still known to contain content that is harmful [23]. Automated content moderation systems are thus both integrated into systems that flag content for human oversight and, ideally, a way to filter out disturbing content before it reaches human eyes.

Many automated content moderation systems focus on identifying hate or toxic speech [8, 21, 24, 42], with additional goals including identifying violent threats [34] or sexual content [9]. Unfortunately, such systems have been found be biased, incorrectly flagging identity-related speech as inappropriate [14, 24, 35]. Identify-focused statements, such as "I'm gay," are often incorrectly flagged [24] because of the frequent use of such identity terms in a negative context online. Algorithm audits, one method for identifying and quantifying biases in algorithmic systems [51], have

shown that human judgments on content dimensions like toxicity may vary between individuals and groups [32, 43], and that human biases in the manual data used to train content moderation systems may be an additional cause of some of these issues [12].

Generative AI systems are increasingly being looked to as creative storytelling devices, with the use of generative AI to create scripts a key issue in the 2023 Hollywood writers' strike [39], and with numerous start-ups using generative AI to author children's books [40]. Yet generative text systems have been shown to produce many types of undesirable content for these contexts, such as overly violent, hateful, or sexual content [28]. To address this, automated content moderation systems are being used as a final filtering step in generative AI systems, where instead of human review, violating content is automatically flagged and removed before potentially problematic text is shown to users [48].

Generative AI content moderation systems have not previously been comprehensively assessed for speech suppression. A first audit of OpenAI's content moderation system found initial evidence of potential biases in the form of overzealous moderation — [46] found widespread flagging of TV episode summaries, with even some PG-rated shows flagged as violations. That audit also indicated that newer versions of OpenAI's large language model (GPT-4) may be incorporating content moderation into the text generation process itself. From the lens of speech generation and potential suppression, this work builds on these efforts via a cross-system audit of identity-related speech suppression.

2.2 Bias audits and generative AI

Large language models have long been known to suffer from gender biases in the underlying embedding space [13, 17]. More recent work has shown that large language models and generative AI systems additionally suffer from: anti-Muslim bias, associating Muslims with violence and terrorism [1, 16]; disability bias, associating disability with negative sentiment [36, 68]; bias against trans and non-binary gender identities, with systems misgendering people and eliciting toxic responses to gender identity statements [57]; anti-LGBTQ bias, with systems more likely to generate an LGBTQ stereotype than a neutral sentence [25]; racial bias that associates 'Asian' with positive sentiment and 'Black' with negative sentiment [16]; and bias against a wide variety of stigmitized groups [49]. Applications suffering from bias when generative AI is used as part of a pipeline include resume screening [5, 44], recommendation letters [38], healthcare [73], housing [45], legal contexts [47], child protective services [26], and emergency services [2].

In the context of storytelling, one recent investigation of four generative AI systems found that when prompted to generate short stories, all of these systems were likely to portray characters and situations representing dominant groups; for example, when asked to write a love story, essentially all generated couples were straight [29]. Another investigation focused on text-to-image generation for children's stories similarly found that the generated images lacked diversity and also that they were too likely to generate sexualized images of women and girls or other biased images to be used for children's content [7]. Previous work in the context of television shows [46] (described above) leaves as an open question whether content moderation practices differently impacted shows related to different identity groups.

2.3 AI safety

The AI safety literature aims to ensure that generative AI systems do not generate text that is undesired across a wide array of potentially concerning use cases (for a survey, see [74]). Risks of these models have been examined and assessed by the companies creating them (e.g., [55]) as well as via external audits and controlled red-teaming efforts (e.g., [63, 64]). Identified risks include the creation of novel chemical weapons [66], dangerous actions by moving robots [61], eliciting a private credit card number and other cybersecurity concerns [64], dangerously incorrect medical misinformation [19], and assisting in disinformation or influence operations [55], along with a wide variety of potential harms in regulated domains [74]. These risks have actualized in real-world harms, including suicide [58], defamation [59], and the creation of child sexual abuse material [15].

To address these concerns, researchers have systematically tried to elicit harmful outputs from LLMs to better understand their risks via red-teaming and other systematic testing [55, 63, 64]. Benchmarks have also been created that identify concerning risks and allow developers to stress test generative models before release (see, e.g., [74]). Such concerns and required testing procedures have gained significant policy attention, including a White House focus on red-teaming [52] and an AI executive order focused on safety [11], a prominent California AI bill [70], and other policy and regulatory efforts that include protections against safety risks of AI systems as a key component [69, 72].

3 DEFINING SPEECH SUPPRESSION

In a traditional content moderation setting, the goal of content moderation is to identify ("flag") inappropriate content to be removed automatically or sent to a human reviewer for further assessment and potential manual removal. Thus, automated content moderation systems are traditionally assessed based on accuracy score, AUC, or other scores that take into account correctness on both true positive (should be flagged) and true negative (should not be flagged) labeled text instances. This has included identity-related work assessing potential bias in such systems [14, 24].

However, in this work we are interested in assessing the potential for *speech suppression*, which we define as incorrectly marking or scoring text as violating content. We will define this term precisely below, but first note the one-sided nature of this goal: we are concerned solely with whether speech identified as a true negative (not violating) is censored, and do not assess systems based on their behavior on text that *should* be flagged as violating. This could be considered a "freedom of speech" goal. As we will describe further in Section 4.1, in addition to supporting our audit aims, this one-sided goal will allow the creation of a new type of dataset useful to the assessment of speech suppression in generative AI content.

In Section 4 we will assess identity-related speech suppression across multiple API return types (both Boolean flags and scores); here, we thus introduce speech suppression measures for both types of outcome. Throughout, our definitional goals focus on a worst-case censorship analysis, aiming to identify if some groups' speech